

Ushtrime ND9a

1.2.149 Në vrimën e vogël, me seksion $S=0.2 \text{ cm}^2$, në fundin e enës rrjedh ujë. Vrima gjendet nën nivelin konstant të ujit prej $h=50 \text{ cm}$. Ç'sasi e ujit rrjedh brenda $t=10 \text{ s}$?

Zgjidhje

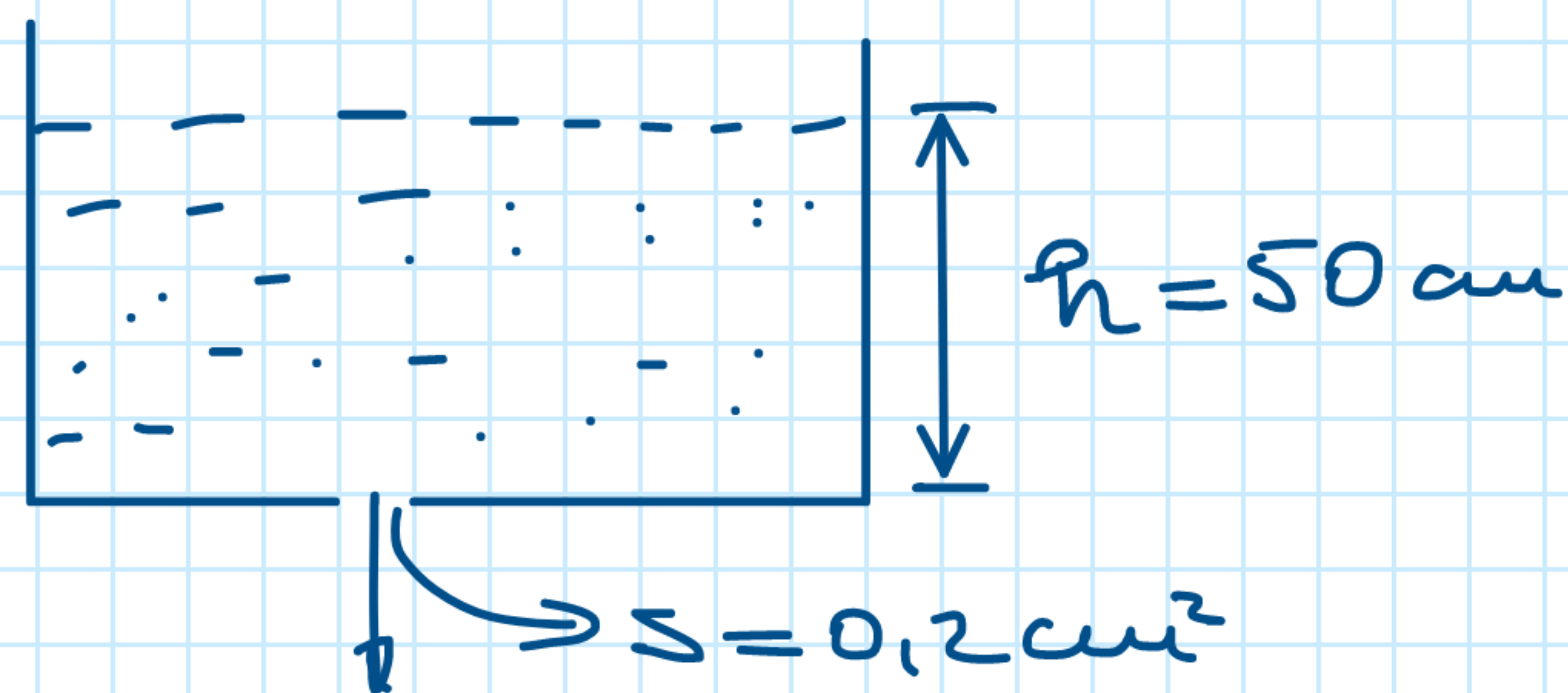
Të dhënat:

$$S = 0.2 \text{ cm}^2 = 0.2 \cdot (10^{-2} \text{ m})^2 = 0.2 \cdot 10^{-4} \text{ m}^2$$

$$1 \text{ cm} = 1 \cdot 10^{-2} \text{ m}$$

$$h = 50 \text{ cm} = 50 \cdot 10^{-2} \text{ m} = 0.5 \text{ m}$$

$$t = 10 \text{ s} \quad g = 9.81 \frac{\text{m}}{\text{s}^2}$$



$$V = ?$$

$$Q = \frac{V}{t} = \frac{S \cdot l}{t} = S \cdot v, \quad \frac{l}{t} = v$$

$$\frac{V}{t} = S \cdot v / t$$

$$V = S \cdot v \cdot t; \quad v = \sqrt{2gh}$$

$$V = S \cdot t \sqrt{2gh}$$

$$V = 0.2 \cdot 10^{-4} \text{ m}^2 \cdot 10 \text{ s} \sqrt{2 \cdot 9.81 \frac{\text{m}}{\text{s}^2} \cdot 0.5 \text{ m}}$$

$$V = 2 \cdot 10^{-4} \text{ m}^2 \cdot \text{s} \sqrt{9.81 \frac{\text{m}^2}{\text{s}^2}}$$

$$V = 2 \cdot 3.13 \cdot 10^{-4} \text{ m}^2 \cdot \cancel{\text{s}} \cdot \frac{\text{m}}{\cancel{\text{s}}} \cdot \sqrt{\cancel{\text{s}^2}}$$

$$V = 6.26 \cdot 10^{-4} \text{ m}^3$$

$$1 \text{ m}^3 = (10^2 \text{ cm})^3 = 1 \cdot 10^6 \text{ cm}^3 \Rightarrow V = 6.26 \cdot 10^{-4} \cdot 10^6 \text{ cm}^3$$

$$V = 6.26 \cdot 10^2 \text{ cm}^3 = \underline{\underline{626 \text{ cm}^3}}$$